

Lida Wang

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🏠 <https://lidawangpsu.github.io/Lida/>

Research Interests

1. Developing statistical models to characterize eQTL architectures at single-cell resolution, enabling a deeper understanding of cell-type- and cell-state-specific gene regulation.
2. Leveraging genetic and electronic health record (EHR) data through statistical and AI-based methods to predict and prevent disease progression.

Appointment

2025–Present **Dana-Farber Cancer Institute and Harvard Medical School**
Postdoctoral Research Fellow, Advisor: **Prof. Alexander Gusev**

Education

2020–2025 **Penn State University**
Ph.D. in Biostatistics, Advisors: **Prof. Dajiang J. Liu**

Summer 2019 **University of Washington**
Statistical Genetics Summer Institute

2018–2019 **Georgetown University**
M.S. in Biostatistics, Advisors: **Prof. Ao Yuan** and **Prof. Ming T. Tan**

Summer 2016 **University of British Columbia**
Mathematics Summer Program

2014–2018 **Southern University of Science and Technology**
B.S. in Mathematics and Applied Mathematics, Advisor: **Prof. Guoliang Tian**

Honors & Awards

2026 Postdoctoral Trainee Research Excellence **Finalist**, American Society of Human Genetics

2025 Reviewer's Choice Abstract, American Society of Human Genetics

2025 U.S. Resource Limited Travel Stipend, American Society of Human Genetics

2024 Reviewer's Choice Abstract, American Society of Human Genetics

2024 Predoctoral Trainee Research Excellence **Semi-Finalist**, American Society of Human Genetics

2024 Vernon M. Chinchilli, PhD Award for Excellence in Biostatistics, Penn State University

2023 Predoctoral Trainee Research Excellence **Finalist**, American Society of Human Genetics

2023 Predoctoral Trainee Awardee, Association of Chinese Geneticists in America

2023 Finkelstein Memorial Trainee Research Award, Penn State University

2019 Statistical Genetics Summer Institutes Scholarship, University of Washington

2018 Biomedical Graduate Tuition Scholarship, Georgetown University

2017 Second Prize in China Mathematical Modeling Competition

2016 Third Prize in Chinese College Students Mathematics Competition

2016 International Exchange Scholarship, SUSTech

2014–2018 Foundation Scholarship for Academic Excellence, SUSTech

Publications

† Co-primary authors; ★ Representative works

Published

- [10] Khunsriraksakul, C., Markus, H., **Wang, L.**, et al. An Integrated Large-Scale Atlas of Protein Quantitative Trait Loci across Olink and SomaScan platforms. medRxiv <https://doi.org/10.1101/2025.10.06.25336803> **ASHG 2024 Postdoctoral Trainee Semi-Finalist, ASHG 2025 Lightning Talk**
- [9] Hu, X., Araujo, D., Khunsriraksakul, C., **Wang, L.**, et al. Multi-Ancestry Transcriptome Prediction with Functionally Informed Variants in TOPMed MESA Improves Performance of Transcriptome-Wide Association Studies. The American Journal of Human Genetics, 113, 828-841 (2026)
- ★[8] **Wang, L.**†, Gao, S.†, et al. Integrating axis quantitative trait loci looks beyond cell types and offers insights into brain-related traits. Nat Commun 16, 10606 (2025). **ASHG 2025 U.S. Resource Limited Travel Stipend and Reviewer's Choice Abstract**
- ★[7] **Wang, L.**†, Markus, H.†, et al. An atlas of single-cell eQTLs dissects autoimmune disease genes and identifies novel drug classes for treatment. Cell Genomics (2025). **ASHG 2023 Plenary Talk and Predoctoral Trainee Finalist; ACGA 2023 Predoctoral Trainee Awardee.**
- [6] Tan, Y., **Wang, L.**, Zhang, H., et al. Interpretable GWAS by linking clinical phenotypes to quantifiable immune repertoire components. Commun Biol 7, 1357 (2024).
- ★[5] **Wang, L.**†, Khunsriraksakul, C.†, et al. Integrating single cell expression quantitative trait loci summary statistics to understand complex trait risk genes. Nat Commun 15, 4260 (2024). **STATGEN 2024 Platform Talk**
- [4] Xu, X., Khunsriraksakul, C., Eales, J., Rubin, S., Scannali, D., Saluja, S., Talavera, D., Markus, H., **Wang, L.**, et al. Genetic imputation of kidney transcriptome, proteome and integrative multi-omics yields new molecular, diagnostic and therapeutic targets for blood pressure and hypertension. Nat Commun 15, 2359 (2024).
- [3] Khunsriraksakul, C., Li, Q., Markus, H., Patrick, M., Sauteraud, R., McGuire, D., Wang, X., Wang, C., **Wang, L.**, et al. Multi-ancestry and multi-trait genome-wide association meta-analyses inform clinical risk prediction and drug repurposing for systemic lupus erythematosus. Nat Commun 14, 668 (2023).
- [2] Khunsriraksakul, C., McGuire, D., Sauteraud, R., Chen, F., Yang, L., **Wang, L.**, et al. Integrating 3D genomic and epigenomic data to enhance target gene discovery and drug repurposing in transcriptome-wide association studies. Nat Commun 13, 3258 (2022). **ASHG 2020 Reviewer's Choice Abstract, PQG 2020 conference honorable mention**
- ★[1] Yuan, A., **Wang, L.**, Tan, M. T. Set-regression with applications to subgroup analysis. Statistics in Medicine. 2022;41(1):180-193.

Presentations

- May. 2026 **Oral Talk, STATGEN 2026, Atlanta, GA**
Single-Cell Gene Expression Heritability Informed by Gene $\times$ Cell-Type/State Interactions
- Jan. 2026 **Seminar, Single Cell and GWAS meeting, Harvard University, Boston, MA**
Single-Cell Gene Expression Heritability Informed by Gene $\times$ Cell-Type/State Interactions
- Oct. 2025 **Poster, ASHG 2025, Boston, MA**
Integrating axis quantitative trait loci looks beyond cell types and offers insights into brain-related traits
- Jun. 2025 **Seminar, Chinese Genomics Meet-up Online (CGM), Virtual**
An atlas of single-cell eQTLs dissects autoimmune disease genes and identifies novel drug classes for treatment.
- Mar. 2025 **Seminar, Genetic Epidemiology Journal Club, Emory University, Virtual**
An atlas of single-cell eQTLs dissects autoimmune disease genes and identifies novel drug classes for treatment.

- Feb. 2025 **Seminar, Dana-Farber Cancer Institute, Boston, MA**
An atlas of single-cell eQTLs dissects autoimmune disease genes and identifies novel drug classes for treatment.
- Jan. 2025 **Seminar, University of Pennsylvania, Philadelphia, PA**
An atlas of single-cell eQTLs dissects autoimmune disease genes and identifies novel drug classes for treatment.
- Nov. 2024 **Poster, ASHG 2024, Denver, CO**
Improved genetic association analysis identifies 29 novel loci influencing progression from preclinical stages to lupus.
- Oct. 2024 **Oral Talk, Multi-Omics Symposium, Hershey, PA**
An atlas of single-cell eQTLs dissects autoimmune disease genes and identifies novel drug classes for treatment.
- Feb. 2024 **Seminar, ACGA Seminar, Virtual**
An atlas of single-cell eQTLs dissects autoimmune disease genes and identifies novel drug classes for treatment.
- Nov. 2023 **Oral Talk, ASHG 2023, Washington, DC**
An atlas of single-cell eQTLs dissects autoimmune disease genes and identifies novel drug classes for treatment.
- Oct. 2023 **Seminar, Penn State University, Hershey, PA**
An atlas of single-cell eQTLs dissects autoimmune disease genes and identifies novel drug classes for treatment.
- Oct. 2023 **Oral Talk, Rheumatology Research Day, Hershey, PA**
Dissecting systemic lupus erythematosus associated genetic loci at single cell resolution.
- Sep. 2023 **Seminar, Penn State University, Hershey, PA**
Genetic analysis of image-derived phenotype in kidney substructures.
- Aug. 2023 **Oral Talk, Calico Life Sciences, South San Francisco, CA**
Genetic analysis of image-derived phenotype in kidney substructures.
- Oct. 2022 **Poster, ASHG 2022, Los Angeles, CA**
Integrating single cell expression quantitative trait loci summary statistics to understand complex trait risk genes.
- Sep. 2022 **Seminar, Penn State University, Hershey, PA**
Integrating single cell expression quantitative trait loci summary statistics to understand complex trait risk genes.
- Aug. 2022 **Oral Talk, SUSTech Statistics and Data Science Seminar, Virtual**
Biostatistics Ph.D. Applications and an Introduction to Statistical Genetics.
- Dec. 2019 **Poster, Research Practicum Presentation, Washington, DC**
Set-regression with applications to subgroup analysis.

Grants

- 2023–2024 **Finkelstein Memorial Trainee Research Award**
Dissecting systemic lupus erythematosus associated genetic loci at single cell level (\$3,000)

Teaching Experience

- Fall 2019 **Teaching Assistant, Statistics Software using R and SAS, Georgetown University**
 Held office hours; graded assignments and exams; prepared solutions; and reviewed exam questions with students.
- Fall 2017 **Teaching Assistant, Probability Theory, SUSTech**
 Held office hours; graded assignments and prepared solutions.

Professional Experience

Journal Reviewer

American Journal of Human Genetics, Genome Biology, Genome Research, Nucleic Acids Research, Biostatistics, Scientific Reports, BMC Cancer, Discover Medicine, etc.

Conference Reviewer

American Society of Human Genetics

Contest Judge

2026 DNA Day Essay Contest by ASHG

Membership

American Society of Human Genetics, American Statistical Association

Industry Experience

Summer 2023 **Calico Life Sciences LLC**, *Research Intern*
Mentor: Dr. Madeleine Cule

Spring 2017 **Shenzhen VITA-COURSE Technology Company**, *Algorithm Engineer Intern*
Mentor: Prof. Heng Peng

References

Prof. Alexander Gusev (Postdoctoral advisor)

Associate Professor of Medicine

Lead, Clinical Computational Oncology Group

Harvard Medical School / Dana-Farber Cancer Institute

Principal Investigator

alexander_gusev@dfci.harvard.edu

Prof. Dajiang Liu (Doctoral advisor)

University Distinguished Professor

Vice Chair for Research, Dept. of Public Health Sciences

Director, Artificial Intelligence and Biomedical Informatics

Co-Chair, Bioinformatics and Genomics Graduate Program

Penn State College of Medicine

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Prof. Laura Carrel

Professor

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